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Roll No

EC-405 (GS)

B.E. IV Semester Examination, June 2020

Grading System (GS)

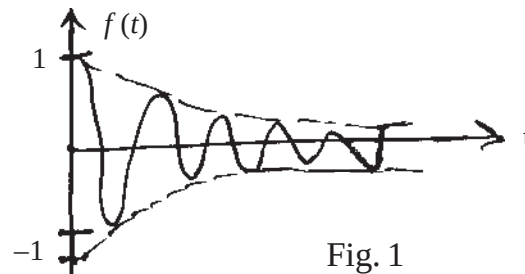
Analog Communication

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Find the Fourier transform of a damped sinusoidal wave form of frequency ω_0 show in fig 1.



2. A 1.0MHz carrier is to be amplitude modulated by music signal with frequencies varying from 50Hz to 20KHz. Find out the range of frequencies of the upper and lower sidebands produced. Hence find the bandwidth of the channel required to transmit this modulated signal.
3. What do you understand by the following signal operations? Explain with examples and mathematical expressions:
i) Time Shifting
ii) Time Scaling
iii) Inversion
4. Explain about sensitivity, selectivity and fidelity of Radio receivers and how can it be improved.
5. If a single tone modulating signal $E_m \cos \omega_m t$. Amplitude modulates a carrier $E_c \cos \omega_c t$. Determine the expression of the AM waveform. Also show the power distribution of the signal and determine the efficiency of the signal.
6. A carrier $A \cos \omega_c t$ is modulated by single tone modulating signal $f(t) = E_m \cos \omega_m t$ find
i) Total modulated power
ii) Root mean square value of the modulated signal and
iii) Transmission efficiency for a 100 percent modulation.

OR

- a) Calculate the figure of merit for AM system.
b) Compare Noise performance of AM and FM.

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7. a) State and explain Paley Wiener criterion. Discuss the causality and physical realizability condition.
b) State and prove Parseval's theorem for power signals.

OR

Obtain the Fourier transform of the signal given below

$$x_1(t) = +1 \text{ for } -2 < t < 0 \text{ and}$$

$$x_2(t) = -1 \text{ for } 0 < t < 2$$

8. Write short notes on any two of the following:
a) Time shift and frequency shift property of Fourier transform.
b) PLL and its use in demodulation
c) Diversity Reception.
